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Inventor(s)	Lee; Jeong-hyeon

Transparent mask equipped with oxygen supply means

Abstract

Proposed is a transparent mask equipped with an oxygen supply means, wherein the transparent mask includes a transparent member configured to be worn on a face and to cover a nose and a mouth, a coupling member configured to enable a periphery of the transparent member to be brought in close contact with the face, an air inflow port provided on one side of the transparent member and configured to allow air to be introduced into an inside of the transparent member, air discharge port provided on an opposite side of the transparent member and configured to allow air to be discharged to an outside of the transparent member, and an air discharge means provided in the air discharge port, wherein the air discharge means includes a discharge flap configured only to allow such that air is discharged to the outside of the transparent member.

Inventors:	Lee; Jeong-hyeon (Jeju-si, KR)
Applicant:	Lee; Jeong-hyeon (Jeju-si, KR)
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Primary Examiner: Patel; Tarla R

Attorney, Agent or Firm: LRK PATENT LAW FIRM

Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates to a transparent mask equipped with an oxygen supply means and, more particularly, to a transparent mask equipped with an oxygen supply means capable of receiving oxygen by connecting a transparent member and an oxygen cylinder as well as allowing a mask wearer's face to be checked through a transparent member.

2. Description of Related Art

(2) In general, a mask is used to block pathogens such as viruses and the like from being transmitted through the respiratory tract or to block harmful dust or fine dust from entering the

human body through the respiratory tract. Regarding such masks, there is a general type in which a mask body made of woven fabric or non-woven fabric and a filter body are configured in a dual structure, and there is a high-functioning type in which an air purifying means and a discharge means are installed in the mask body.

(3) On the other hand, the general type of mask has the advantage of being inexpensive, but there are disadvantages in that when a wearer inhales and exhales, air needs to pass through both the mask body and the filter body, so it is difficult to breathe because the air does not flow smoothly, and moisture is easily formed on the inside of the mask.

(4) The high-functioning type mask has advantages of not causing breathing difficulty and not allowing moisture to be easily formed because the air is circulated smoothly because suction and discharge are separate due to the air purification means and the discharge means but has the disadvantage of being expensive.

(5) However, since the mask covers the mask wearer's face (nose part and mouth part), there is a problem in that the expression of the mask wearer's face may not be seen, causing a sense of caution or rejection rather than a sense of familiarity.

(6) In addition, in the case of deaf people who may not hear sounds due to hearing impairment, there is a problem in that communication may not be accomplished because the shape of the mouth may not be observed.

(7) In order to solve the above problems, film-type transparent mask has been recently registered as Korean Patent No. 10-1681100. This related art is a film type transparent mask, which includes a mask body that is in close contact with all of or a part of the user's face, a filter part provided on a lower surface of the mask body to purify the sucked air, and a fixing part configured to fix the mask to the face, wherein the filter part includes a first film in which one or more vent holes are repeatedly provided at predetermined intervals, a second film in which one or more vent holes are repeatedly provided at predetermined intervals, and a filter layer provided between the first film and the second film, wherein The first vent holes of the first film and the second vent holes of the second film are disposed to be staggered with each other, and the filter layer is a nonwoven fabric filter or a nanoweb filter. In addition, an airtight part made of hydrogel, adhesive, polyethylene foam, acrylic foam, or rubber material that may be brought into close contact with the user's face is provided on the edge of the film type transparent mask, flow paths of various patterns are provided between a plurality of vent holes in the first film and the second film so as to maximize filtration paths of the sucked air by having a length longer than a linear distance between adjacent vent holes, and an inner surface of the flow path of the various patterns is configured to have adhesive or adsorbent for adsorbing contaminants applied thereto during passing of the air containing external harmful substances through the flow path.

(8) In the related art above, the mask body is composed of a transparent plastic film, and thus it is possible to solve the problems of conventional art, but there is a problem that it is difficult to use in an underground space and the inside of a large water tank, where a breathing environment is poor due to ventilation difficulty, or a tank for storing liquid.

(9) In addition, when a fire occurs, there is a problem that breathing of the mask wearer becomes difficult because toxic gases may not be filtered.

(10) The foregoing is intended merely to aid in the understanding of the background of the present disclosure and is not intended to mean that the present disclosure falls within the purview of the related art that is already known to those skilled in the art.

DOCUMENTS OF RELATED ART

Patent Document

(11) (Patent Document 0001) Korean Patent No. 10-1681100

SUMMARY OF THE INVENTION

(12) Accordingly, the present disclosure has been made keeping in mind the above problems occurring in the related art, and the present disclosure is intended to provide a transparent mask

equipped with an oxygen supply means capable of allowing a mask wearer's face to be checked through a transparent member as well as enabling smooth breathing.

(13) In addition, the present disclosure is intended to provide the transparent mask equipped with an oxygen supply means capable of receiving oxygen by connecting an oxygen cylinder and/or a second air inflow means to a transparent member.

(14) In addition, the present disclosure is intended to provide the transparent mask equipped with an oxygen supply means capable of receiving oxygen and external air even under water or in an environment difficult to ventilate by connecting the oxygen cylinder and/or the second air inflow means to the transparent member.

(15) In order to achieve the above objectives, according to one aspect of the present disclosure, there may be provided a transparent mask equipped with an oxygen supply means, the transparent mask including: a transparent member configured to be worn on a face and to cover a nose and a mouth; a coupling member configured to enable a periphery of the transparent member to be brought in close contact with the face; an air inflow port provided on one side of the transparent member and configured to allow air to be introduced into an inside of the transparent member; air discharge port provided on an opposite side of the transparent member and configured to allow air to be discharged to an outside of the transparent member; and an air discharge means provided in the air discharge port, wherein the air discharge means includes a discharge flap configured only to allow such that air is discharged to the outside of the transparent member.

(16) The transparent mask may further include an air inflow means provided in the air inflow port, wherein the air inflow means includes: an inflow flap configured only to allow such that air is introduced into the inside of the transparent member; and a filter configured to filter the air introduced thereinto.

(17) The filter may include carbon powder and copper powder.

(18) The filter may have a carbon filter layer containing carbon powder and a copper filter layer containing copper powder installed to be overlapped, and the carbon filter layer and the copper filter layer may be configured to be added or replaced.

(19) The transparent mask may further include an air inflow pipe with a flexible material, the pipe having one side coupled to the air inflow port.

(20) The air inflow pipe may have an opposite side branched off into a first branch pipe and a second branch pipe, wherein the first branch pipe may be provided with a coupling assembly to which an oxygen cylinder is coupled, whereby oxygen may be supplied thereto from the oxygen cylinder, and the second branch pipe may be provided with a second air inflow means, whereby external air may be supplied thereto, wherein a second air inflow means may include: a second inflow flap configured only to allow such that air is introduced into the inside of the transparent member; and a second filter configured to filter the air introduced in.

(21) The transparent mask may further include an oxygen supply amount control part provided in the coupling assembly, wherein the oxygen supply amount control part may control an amount of oxygen supplied from the oxygen tank as a control valve rotates.

(22) The second branch pipe may be provided long and may be provided with a floating body floatable on water on an end portion thereof, and the second air inflow means may be exposed into air over the floating body, thereby allowing fresh air above the water to be introduced thereinto.

(23) The second branch pipe may have an end portion provided long, wherein the end portion may be provided with a weight that may fall down to a lower side due to a gravitational action or with a ring, whereby the second air inflow means may allow fresh air outside a window or a workplace to be introduced into the second branch pipe when fire occurs by being positioned at an appropriate place outside the window of a building or outside the workplace having a space provided therein.

(24) The oxygen supply amount control part may include: an opening/closing hole penetrating through a coupling pipe so as to communicate with an air movement hole; an opening/closing ball mounted in the opening/closing hole and configured to open and close the air movement hole; a

spring mounted on the coupling pipe; a control valve mounted to be able to rotate to the coupling pipe while compressing the spring and configured to control pressing force of the opening/closing ball while rotating around the coupling pipe; and a fixing ring mounted on the coupling pipe so as to prevent the control valve from being separated.

(25) According to the present disclosure, it is possible to check a face of a mask wearer through the transparent member, as well as to enable the mask wearer to smoothly breathe.

(26) In addition, oxygen can be supplied by connecting an oxygen cylinder and/or a second air inflow means to a transparent member.

(27) In addition, by connecting the oxygen cylinder and/or the second air inflow means to the transparent member, oxygen and external air can be supplied under water or in an environment where ventilation is difficult.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other objectives, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description when taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a perspective view of a transparent mask of the present disclosure;

(3) FIG. 2 is an exploded perspective view showing an air discharge means of the present disclosure;

(4) FIG. 3 is an exploded perspective view showing the air inflow means of the present disclosure;

(5) FIG. 4 is a sectional view of FIG. 3;

(6) FIG. 5 is a perspective view of a transparent mask according to another embodiment of the present disclosure;

(7) FIG. 6 is an enlarged exploded perspective view of FIG. 5

(8) FIG. 7 is an exploded perspective view showing a second air inflow means of the present disclosure;

(9) FIG. 8 is a sectional view taken along line A-A' of FIG. 5; and

(10) FIG. 9 is an operating state diagram of an oxygen supply amount control part of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

(11) Hereinafter, some embodiments of the present disclosure will be described in detail with reference to exemplary drawings. In adding reference numerals to the components of each drawing, the same components are given the same reference numerals as much as possible even though indicated on different drawings. In addition, in describing the embodiment of the present disclosure, when it is determined that a detailed description of a related known configuration or function interferes with the understanding of the embodiment of the present disclosure, the detailed description thereof will be omitted.

(12) In addition, in describing components of the embodiment of the present disclosure, terms such as first, second, A, B, (a), (b), and the like may be used. Such terms are only for distinguishing components from other elements, and the essence, order, or order of the components are not limited by the terms. When it is described that a component is “connected”, “coupled”, or “combined” with another component, the component may be directly connected, “coupled”, or “combined” with another component, but another component may be “connected”, “coupled”, or “combined” between each component.

(13) Hereinafter, a transparent mask having an oxygen supply means according to an example of the present disclosure will be described with reference to the drawings.

(14) FIG. 1 is a perspective view of a transparent mask equipped with an oxygen supply means

according to an example of the present disclosure.

(15) The transparent mask equipped with an oxygen supply means according to an example of the present disclosure may include a transparent member **110**, a coupling member **110'**, an air inflow port **120**, an air discharge port **120'**, and an air outlet means **130**.

(16) The transparent member **110** is worn on the face to cover the nose and mouth and may be provided with a transparent material. The material of the transparent member **110** may be polyethylene (PE) or the like. The transparent member **110** may be manufactured in a shape convenient to wear taking the nose, mouth, face, and the like into consideration. The transparent member **110** may be bent in response to the size or shape of the wearer's face when worn but may need to stay to an extent not to wobble due to exhalation or inhalation. Silicone, rubber, and the like may be deformed and brought into close contact with the wearer's face to prevent air from infiltrating in or leaking out.

(17) The coupling member **110'** configured to enable a periphery of the transparent member **110**, that is, silicon **111**, rubber, and the like to be brought in close contact with the face and may be a band or earpiece connected to opposite sides of the left and right of the transparent member **110** and the like.

(18) The air inflow port **120** is a hole through which air is introduced into the transparent member **110** when the mask wearer breathes in and may be provided passing through at one side of the transparent member **110**. One side on which the air inflow port **120** is installed may be an upper side, a lower side, a left side, a right side, or the like of the transparent member **110**.

(19) The air discharge port **120'** is a hole through which air is discharged to the outside of the transparent member **110** when the mask wearer exhales and may be provided passing through at an opposite side of the transparent member **110**. The opposite side corresponds to a side and refers to a part other than the one side. That is, the opposite side may be a position opposite to the one side or adjacent to the one side.

(20) The one side and the opposite side of the transparent member **110** may be positioned so that the mask wearer's mouth shape may be seen so as to allow the hearing impaired to identify the wearer's mouth shape, that is, the speaking shape.

(21) With reference to FIG. 2, the transparent mask may further include an air discharge means **130** provided in the air discharge port **120'**. The air discharge means **130** may include an air discharge guide member **131**, a discharge flap **132**, and an air discharge fastening member **133**.

(22) The air discharge guide member **131** is a cylindrical member and may be attached to the outer surface of the transparent member **110** by attachment, fusion, and the like or may be provided integrally with the transparent member **110**. A screw may be provided, or a latching portion may be provided, on an outer circumferential surface of the air discharge guide member **131**. The inner circumferential surface of the air discharge guide member **131** may be provided with a radially shaped discharge movement guide protrusion **131'** thereon or may be provided without the discharge movement guide protrusion **131'** thereon. A hole may be provided by passing through the inside of the air discharge guide member **131**, that is, a surface brought in contact with the transparent member **110**, entirely, or by passing through only a portion corresponding to the air discharge port **120'**.

(23) The discharge flap **132** opens and closes the air discharge port **120'** and may be provided with a soft synthetic resin material. In a state in which only one side is attached, the discharge flap **132** may be bent during exhalation so that the air discharge port **120'** is opened and may block the air discharge port **120'** by returning back during inhalation. In addition, in a completely separated state, the discharge flap is moved during inhalation and may close the air discharge hole, thereby blocking the inflow of air, and is moved during exhalation and may open the air discharge hole, thereby discharging the air. The discharge movement guide protrusion **131'** may facilitate movement of the discharge flap **132** by minimizing frictional force. The discharge flap **132** may be provided in a sawtooth shape in which a portion corresponding to the discharge movement guide

protrusion **131'** is cut off. That is, the discharge flap **132** may be provided by being cut off to allow to an extent only a portion of the discharge movement guide protrusion **131'** to be inserted therein during discharge. Such a sawtooth shape may allow the discharge flap **132** to move in a stable state without being rotated, thereby perfectly performing opening and closing of the air discharge port **120'**.

(24) An air discharge fastening member **133** may be fastened and fixed to an air discharge guide member **131** by a screw or a fitting method, thereby preventing the discharge flap **132** from being separated. When damaged or aged, the discharge flap **132** may be replaced with a new one after disassembling the air discharge fastening member **133**.

(25) With reference to FIGS. **3** and **4**, the transparent mask may further include an air inflow means **140** provided in the air inflow port **120**. The air inflow means **140** may include an air inflow guide member **141**, an inflow flap **142**, an installation guide member **143**, a filter **144**, and an air inflow fastening member **145**.

(26) The air inflow guide member **141** is a cylindrical member and may be coupled to the outer surface of the transparent member **110** by attachment, fusion, and the like or may be provided integrally with the transparent member **110**. On the outer peripheral surface of the air inflow guide member **141**, a screw may be provided, or a latching portion may be provided. An inner circumferential surface of the air inflow guide member **141** may be provided with an inflow guide protrusion **141'** radially shaped thereon or may be provided without the inflow guide protrusion **141'** thereon. A hole may be provided by passing through the inside of the air inflow guide member **141**, that is, the surface brought in contact with the transparent member **110**, entirely, or by passing through only a portion corresponding to the air inflow port **120**. The inflow movement guide protrusion **141'** may be provided in two stages or may be provided in one stage on the inner side, that is, on a side toward the air inflow port **120**. An inflow flap **142** to be described later may be located on the outer side of the inflow movement guide protrusion **141'** of the inner side, that is, a position of the inflow movement guide protrusion **141'** of the outer side.

(27) The inflow flap **142** may be provided with a soft synthetic resin and may open and close an inflow through hole **143'** of an installation guide member **143** to be described later while moving along the inflow movement guide protrusion **141'** of an outer side. In addition, the inflow flap **142** having one side attached to the installation guide member **143** may introduce air by being bent to be spaced apart from the inflow through hole **143'** during inhalation and may close the inflow through hole **143'** by being returned during exhalation. In addition, in an unattached state, the inflow flap **142** is moved during inhalation and may open the inflow through hole **143'** and is moved and may close the inflow through hole **143'** during exhalation. The inflow flap **142** may be provided in a sawtooth shape in which a portion in which the inflow movement guide protrusion **141'** of an outer side is located is cut off. That is, the inflow flap **142** may be provided with a cut off portion into which a portion of the inflow movement guide protrusion **141'** of an outer side is inserted. Such a sawtooth shape may allow the inflow flap **142** to move in a stable state without being rotated, thereby allowing opening and closing of the inflow through hole **143'** to be perfectly performed.

(28) The installation guide member **143** is a cylindrical shape, an outer side and an inner side are open, an intermediate membrane having the inflow through hole **143'** in the center is provided, and a first space and a second space may be provided on the inner side and the outer side, respectively. The installation guide member **143** may be coupled to the air inflow guide member **141** by having a screw or a latching portion provided on the inner circumferential surface of an inner side. The installation guide member **143** may have a screw or a latching portion provided on the outer circumferential surface. The inner peripheral surface of the first space is fastened to the air inflow guide member **141** and is brought into close contact with the same, thereby preventing the inflow flap **142** from being separated. The second space may accommodate a filter **144** to be described later.

(29) The filter **144** filters foreign substances contained in the incoming air and may be installed in the second space of the installation guide member **143**. The filter may include carbon powder and copper powder. In the filter, a carbon filter layer containing carbon powder on a nonwoven fabric, woven fabric, sponge, and the like, and a copper filter layer containing copper powder on a nonwoven fabric, woven fabric, sponge, or the like may be overlapped. The carbon filter layer and the copper filter layer may be added or replaced.

(30) The air inflow fastening member **145** is fastened to the installation guide member **143** to fix the filter **144**.

(31) With reference to FIG. 5, the transparent mask may further include an air inflow pipe **150** having one side coupled to the air inflow port **120**. The air inflow pipe **150** may be provided with a flexible material. The air inflow pipe **150** may be directly coupled to the transparent member **110** by attachment, fusion, or the like, or coupled to the air inflow guide member **141** of the air inflow means **140**. An opposite side of the air inflow pipe **150** may include a first branch pipe **151** and a second branch pipe **152** that are branched off. The first branch pipe **151** is provided with a coupling assembly **170** to which an oxygen cylinder **160** is coupled so that oxygen may be supplied from the oxygen cylinder **160**. The second branch pipe **152** may be provided with a second air inflow means **180**.

(32) With reference to FIG. 6, the coupling assembly **170** may include a coupling member **171** and a coupling pipe **172** with a configuration to be able to detach from the oxygen cylinder **160**.

(33) The coupling member **171** may be mounted on the oxygen discharge part **161** of the oxygen cylinder **160**. Wings coupled to wing grooves of the oxygen cylinder **160** may be provided at end portions under the coupling member **171**. The wings are provided as a pair and are coupled to the oxygen cylinder **160** to maintain bonding force with the oxygen cylinder.

(34) The coupling tube **172** has a tubular shape, and an air movement hole **173** may be provided therein. The injection nozzle **162** of the oxygen cylinder **160** may be coupled to an inner end portion of the air movement hole **173**.

(35) With reference to FIG. 7, the second air inflow means **180** may include a connection port **181**, a second inflow flap **182**, a second installation guide member **183**, a second filter **184**, and a second air inflow fastening member **185**. The second air inflow means **180** is similar to the above-described air inflow means **140**. Therefore, omitted parts may refer to the air inflow means **140**.

(36) The connection port **181** of the second air inflow means **180** may be coupled by being fitted, attached, bonded, or fusion-bonded to the second branch pipe **152** or integrally provided with the second branch pipe **152**, and a screw or a latching portion may be provided on an outer circumferential surface thereof. The connection port **181** is a cylindrical member, and an inner circumferential surface of the connection port **181** may be provided with a second inflow movement guide protrusion **181'** radially shaped thereon or may be provided without the second inflow inner protrusion **181'**. A hole may be provided by passing through only a portion of the inner side of the connection port **181**, that is, a surface being brought into contact with the second branch pipe **152**. The second inflow inner protrusion **181'** may be provided in two stages or may be provided in one stage on an inner side of the connection port **181**, that is, the side of the second branch tube **152**. A second inflow flap **182** to be described later may be located on an outer side of the second inflow movement guide protrusion **181'** of an inner side.

(37) The second inflow flap **182** is made of a soft synthetic resin and may open and close a second inflow through hole **183'** of a second installation guide member **183** while moving along the second inflow movement guide protrusion **181'** of the outer side. In addition, attached to one side of the second installation guide member **183** to be described later, the second inflow flap **182** may introduce air by being bent to be spaced apart from the second inflow through hole **183'** during inhalation and close the second inflow through hole **183'** by being returned during exhalation. When not attached, the second inflow flap **182** is moved along the second inflow inner protrusion **181'** of the outer side during inhalation and may open the second inflow through hole **183'** and is

moved along the second inflow inner protrusion **181'** during exhalation and may close the second inflow through hole **183'**. The second inflow flap **182** may be provided in a sawtooth shape in which a portion in which the second inflow inner protrusion **181'** of the outer side is located is cut off. Such a sawtooth shape may allow the second inflow flap **182** to move in a stable state without being rotated, thereby perfectly performing opening and closing of the second inflow through hole **183'**.

(38) The second installation guide member **183** is a cylindrical shape, an outer side and an inner side are open, an intermediate membrane having the second inflow through hole **183'** in the center is provided, and a third space and a fourth space may be provided on the inner side and the outer side, respectively. The second installation guide member **183** may be coupled to the connection port **181** by having a screw or a latching portion provided on the inner circumferential surface of an inner side. The second installation guide member **183** may have a screw or a latching portion provided on the outer circumferential surface. The inner peripheral surface of the third space is fastened to the connection port **181** and is brought into close contact with the same, thereby preventing the second inflow flap **182** from being separated. The fourth space may accommodate a second filter **184** to be described later.

(39) The second filter **184** filters foreign substances contained in the incoming air and may be installed in the fourth space of the second installation guide member **183**. The second filter **184** may include carbon powder and copper powder. In the second filter **184**, a carbon filter layer containing carbon powder on a nonwoven fabric, woven fabric, sponge, and the like, and a copper filter layer containing copper powder on a nonwoven fabric, woven fabric, sponge, or the like may be overlapped. The carbon filter layer and the copper filter layer may be added or replaced.

(40) The second air inflow fastening member **185** is fastened to the second installation guide member **183** to fix the second filter **184**.

(41) In addition, with reference to FIG. 6, the transparent mask further includes an oxygen supply amount control part **190** provided in the coupling assembly **170**, wherein the oxygen supply amount control part **190** may include an opening/closing hole **191**, an opening/closing ball **192**, a spring **193**, a control valve **194**, and a fixing ring **195**.

(42) The opening/closing hole **191** may be a hole penetrating through the coupling pipe **172** so as to communicate with the air movement hole **173**.

(43) The opening/closing ball **192** is mounted in the opening/closing hole **191** to open and close large or small the air movement hole **173** according to the rotation of the control valve **194**.

(44) The spring **193** may elastically support the control valve **194** mounted on the coupling pipe **172**.

(45) The control valve **194** may be mounted to be able to rotate to the coupling pipe **172** while compressing the spring **193**. The control valve **194** rotates around the coupling pipe **172** and may control pressing force of the opening/closing ball **192** with an inner spiral-shaped control surface.

(46) A fixing ring **195** may be mounted on the coupling pipe **172**, whereby the control valve **194** may be prevented from being separated, and a control surface of the control valve **194** and the opening/closing ball **192** are to be located on the same surface.

(47) A pair of rotation control members **196** configured to control a rotation angle of the control valve **194** may be provided in the coupling member **171**.

(48) In addition, the transparent mask may be provided with a floating body installed on the second branch pipe **152**. The floating body floats on water and may be STYROFOAM® (Extended Polystyrene Foam, XPS), a tube, or the like. The floating body is provided with the second air inflow means **180** exposed into the air over the floating body, so that fresh air above the water may be introduced thereinto. The floating body may allow an end part of the second branch pipe **152** to float on the water, thereby allowing a person wearing the transparent mask to dive into a river, sea, water tank, or the like. In this case, the oxygen cylinder may be installed on the body or waist of the wearer.

(49) In addition, the transparent mask may be provided with a weight installed on the second branch pipe **152**. Due to a gravitational action of the weight, the second air inflow **180** may descend outside a window of a building, thereby allowing fresh air from the outside to be introduced into the transparent mask in case of a fire.

(50) In a normal case, that is, the case without the floating body or weight, the present embodiment may be easily used by patients, the elderly, and the like who require a lot of oxygen.

(51) The following describes the process of using one embodiment of the present disclosure.

(52) First, the transparent mask **100** is brought into close contact with a face of a mask wearer. In this case, the transparent mask **100** may cover the nose and mouth of the mask wearer, and the silicone **111** attached to the inner circumference of the transparent member **110** may be brought into close contact with the face.

(53) While the transparent mask **100** is brought into close contact with the mask wearer's face, the transparent mask **100** may be worn by allowing the coupling member **110'** to be positioned on a mask wearer's head or to be hung on the mask wearer's ears. At this time, the silicone **111** and the like attached to the inner circumference of the transparent member **110** by the coupling member **110'** is deformed and brought into close contact with the mask wearer's face, thereby preventing air from infiltrating in or leaking out.

(54) When the mask wearer wearing the transparent mask **100** breathes in, by the inhaling suction force, the discharge flip **132** of the air discharge means **130** closes the air discharge port **120'** while moving to an inner side, that is, in a direction of the transparent member **110** along the discharge movement guide protrusion **131'**. Accordingly, it is possible to prevent the inflow of external air into the inside of the transparent member **110**.

(55) In addition, by the inhaling suction force, the inflow flap **142** of the air inflow means **140** opens the inflow through hole **143'** of the installation guide member **143** while moving to the inner side, that is, in a direction of the transparent member **110** along the inflow movement guide protrusion **141'** of the air inflow guide member **141**. Accordingly, external air is introduced into the transparent member **110** through the air inflow port **120**, and the mask wearer may breathe with the air introduced thereinto.

(56) At this time, the inflow movement inner protrusion **141'** is provided in two stages, whereby the air inflow port **120** is not allowed to be closed even when the inflow flap **142** moves in the inward direction, thereby being allowed to guide the inflow of external air. In addition, since the external air flows in through the filter **144**, it is possible to filter fine dust and bacteria included in the external air.

(57) On the other hand, when the mask wearer breathes out, by the exhaling discharge force, the inflow flap **142** of the air inflow means **140** closes the inflow through hole **143'** of the installation guide member **143** while moving to an outer side, that is, in an opposite direction to the member **110** along the inflow movement guide protrusion **141'** of the air inflow guide member **141**. Accordingly, it is possible to prevent the internal air of the transparent member **110** from being discharged.

(58) In addition, by the exhaling discharge force, the discharge flip **132** of the air discharging means **130** opens the air discharge port **120'** while moving to the outer side, that is, in the opposite direction to the member **110** along the discharge movement guide protrusion **131'**. Accordingly, air inside the transparent member **110** may be discharged.

(59) While repeating the above process, the mask wearer may breathe in and breathe out even while wearing the transparent mask **100**, thereby enabling smooth breathing and preventing dust or bacteria from being infiltrated in. In addition, since the shape of the mask wearer's mouth may be checked due to the transparent member **110**, even the hearing impaired may identify the contents.

(60) In addition, to describe the use process of another embodiment of the present disclosure, the transparent mask **100** is brought into close contact with a face of a mask wearer. At this time, the transparent mask **100** may cover the nose and mouth of the mask wearer, and the silicone **111**

attached to the inner circumference of the transparent member **110** may be brought into close contact with the face.

(61) While the transparent mask **100** is brought into close contact with the mask wearer's face, the transparent mask **100** may be worn by allowing the coupling member **110'** to be positioned on a mask wearer's head or to be hung on the mask wearer's ears. At this time, the silicone **111** attached to the inner circumference of the transparent member **110** by the coupling member **110'** is deformed and brought into close contact with the mask wearer's face, thereby preventing air from infiltrating in or leaking out.

(62) When the mask wearer wearing the transparent mask **100** breathes in, by the inhaling suction force, the discharge flip **132** of the air discharge means **130** closes the air discharge port **120'** while moving to an inner side, that is, in a direction of the transparent member **110** along the discharge movement guide inner protrusion **131'**. Accordingly, it is possible to prevent the inflow of external air into the inside of the transparent member **110**.

(63) In addition, by the inhaling suction force, the second inflow flap **182** of the second air inflow means **180** opens the second inflow through hole **183'** of the second installation guide member **183** while moving to the inner side, that is, in a direction of the transparent member **110** along the second inflow inner protrusion **181'** of the connection port **181**. Accordingly, external air is introduced into the inside of the transparent member **110** through the air inflow port **120** via the second branch pipe **152** branched off from the air inflow pipe **150**, and the mask wearer may breathe with the air introduced thereinto.

(64) At this time, the second inflow movement guide protrusion **181'** is provided in two stages, whereby the second inflow flap **182** is unable to close the inside of the connection port **181** even when moved in the inward direction, thereby being allowed to guide the inflow of external air. In addition, since the external air flows in through the second filter **184**, it is possible to filter fine dust and bacteria included in the external air.

(65) On the other hand, when the mask wearer breathes out, by the exhaling discharge force, the second inflow flap **182** of the second air inflow means **180** closes the second inflow through hole **183'** of the second installation guide member **183**, while moving to an outer side, that is, in an opposite direction to the member **110** along the second inflow inner protrusion **181'** of the second air inflow guide member **181**. Accordingly, it is possible to prevent the internal air of the transparent member **110** from being discharged.

(66) In addition, by the exhaling discharge force, the discharge flip **132** of the air discharging means **130** opens the air discharge port **120'** while moving to the outer side, that is, in the opposite direction to the member **110** along the discharge movement guide protrusion **131'**. Accordingly, air inside the transparent member **110** may be discharged.

(67) While repeating the above process, the mask wearer may breathe in and breathe out even while wearing the transparent mask **100**, thereby enabling smooth breathing. The second air inflow means **80** may be provided with the second branch pipe **152** securing a certain length, whereby the air may be easily supplied thereto by positioning the second branch pipe **152** on an upper side of the water surface, at an outside of a workplace with a poor breathing environment, for example, a tank lorry, or at an outside of the window. To this end, the second air inflow means **180** may be provided with a floating body that may float on water, a weight that falls down to a lower side due to a gravitational action, a ring that may be fixed where necessary, or the like.

(68) In addition, with reference to FIG. **9**, the coupling assembly **170** may be connected to the first branch pipe **151** branched off from the air inflow: pipe **150**, and an oxygen cylinder **160** may also be installed in the coupling assembly **170** to be used. For the oxygen stored in the oxygen cylinder **160**, when the control valve **194** of the oxygen supply amount control part **190** is rotated in forward and reverse directions, the curved control surface inside the control valve **194** presses the opening/closing ball **192**, and the opening/closing ball **192** may control the oxygen supply amount by controlling the open area of the air movement hole **173** of the coupling assembly **170** while

moving along the opening/closing hole **191**.

(69) In addition, the opening/closing ball **192** may also control the oxygen supply amount by pressing the opening/closing member coupled to the inside of the air movement hole **173** to control the opening area of the through hole inside the opening/closing member.

(70) The forward and reverse rotation angles of the control valve **194** may be restricted by rotation control members **196** provided as a pair in the coupling member **171**.

(71) Oxygen supplied according to the control of the oxygen supply amount control part **190** is supplied to the inside of the transparent member **110** through the air inflow port **120** via the first branch pipe **151** and the air inflow pipe **150**, and the mask wearer may breathe with the oxygen supplied into the inside of the transparent member **110**.

(72) As described above, the second air inflow means **180** and/or the oxygen cylinder **160** is connected to the transparent member **110**, so it is possible not only to breathe by supplying external air and/or oxygen for a certain period of time in case of fire but also to prevent damage such as suffocation by toxic gas, poisoning by poisonous gas, and the like. In addition, the work environment may be improved because it is possible to breathe by receiving external air and oxygen even in a workplace with a poor breathing environment as well as to be able to work while staying in the workplace for a long time.

(73) Since the filters contain carbon powder and copper powder, fine dust filtration and antibacterial, sterilizing, and deodorizing effects may be obtained, and after a certain period of time, additional filters may be added or replaced.

(74) In the above, even though it has been described that all components constituting the embodiment of the present disclosure are combined or operated in combination, the present disclosure is not necessarily limited to this embodiment. That is, within the scope of the present disclosure, all the components may also be configured or operated by selectively combining one or more components. In addition, terms such as “include”, “compose”, or “have” described above mean that the corresponding component may be immanent unless otherwise stated, so it should be construed as being able to include other components further rather than excluding other components. All terms including technical and scientific terms have the same meaning as commonly understood by one of ordinary skill in the art to which the present disclosure belongs unless otherwise defined. Terms commonly used, such as those defined in the dictionary, should be interpreted as being consistent with the contextual meaning of the related art and are not interpreted in an ideal or excessively formal meaning unless explicitly defined in the present disclosure.

(75) The above description is merely illustrative of the technical spirit of the present disclosure, and various modifications and variations will be possible without departing from the essential characteristics of the present disclosure by those skilled in the art to which the present disclosure pertains. Therefore, the embodiments disclosed in the present disclosure are not intended to limit the technical spirit of the present disclosure but to explain, and the scope of the technical spirit of the present disclosure is not limited by these embodiments. The protection scope of the present disclosure should be interpreted by the following claims, and all technical ideas within the scope equivalent thereto should be construed as being included in the scope of the present disclosure.

Claims

1. A transparent mask equipped with an oxygen supply means, the transparent mask comprising: a transparent member configured to be worn on a face and to cover a nose and a mouth; a coupling member configured to enable a periphery of the transparent member to be brought in close contact with the face; an air inflow port provided on one side of the transparent member and configured to allow air to be introduced into an inside of the transparent member; an air discharge port provided on an opposite side of the transparent member and configured to allow air to be discharged to an outside of the transparent member; an air discharge means provided in the air discharge port; and

an air inflow pipe with a flexible material, the pipe having one side coupled to the air inflow port, wherein the air discharge means includes a discharge flap configured only to allow such that air is discharged to the outside of the transparent member; wherein the air inflow pipe has an opposite side branched off into a first branch pipe and a second branch pipe, wherein the first branch pipe is provided with a coupling assembly to which an oxygen cylinder is coupled, whereby oxygen is supplied thereto from the oxygen cylinder, and the second branch pipe is provided with a second air inflow means, whereby external air is supplied thereto, wherein a second air inflow means includes: a second inflow flap configured only to allow such that air is introduced into the inside of the transparent member; and a second filter configured to filter the air introduced in.

2. The transparent mask of claim 1, further comprising: an air inflow means provided in the air inflow port, wherein the air inflow means includes: an inflow flap configured only to allow such that air is introduced into the inside of the transparent member; and a filter configured to filter the air introduced thereinto.

3. The transparent mask of claim 2, wherein the filter comprises carbon powder and copper powder.

4. The transparent mask of claim 2, wherein the filter has a carbon filter layer containing carbon powder and a copper filter layer containing copper powder installed to be overlapped, and the carbon filter layer and the copper filter layer are configured to be added or replaced.

5. The transparent mask of claim 1, further comprising: an oxygen supply amount control part provided in the coupling assembly, wherein the oxygen supply amount control part controls an amount of oxygen supplied from the oxygen tank as a control valve rotates.

6. The transparent mask of claim 1, wherein the second branch pipe is provided long and is provided with a floating body floatable on water on an end portion thereof, and the second air inflow means is exposed into air over the floating body, thereby allowing fresh air above the water to be introduced thereinto.

7. The transparent mask of claim 1, wherein the second branch pipe has an end portion provided long, wherein the end portion is provided with a weight that falls down to a lower side due to a gravitational action or with a ring, whereby the second air inflow means allows fresh air outside a window or a workplace to be introduced into the second branch pipe when fire occurs by being positioned at an appropriate place outside the window of a building or outside the workplace having a space provided therein.

8. The transparent mask of claim 5, wherein the oxygen supply amount control part comprises: an opening/closing hole penetrating through a coupling pipe so as to communicate with an air movement hole; an opening/closing ball mounted in the opening/closing hole and configured to open and close the air movement hole; a spring mounted on the coupling pipe; a control valve mounted to be able to rotate to the coupling pipe while compressing the spring and configured to control pressing force of the opening/closing ball while rotating around the coupling pipe; and a fixing ring mounted on the coupling pipe so as to prevent the control valve from being separated.
